

Zos versus Unix/Linux AN introduction

DB2 product editions available for z/OS and Linux, UNIX, Windows (LUW) are so different that DB2 database administrators must often possess different skills to deliver on different platforms. This article will help you get started with DB2 that is available on the other platform. It assumes that you would supplement it with required knowledge of respective hardware and operating systems. This article aims to introduce different techniques in handling database administration duties across platforms and should not be used to compare the two products for their features or capabilities.

DB2 LUW is also available in various editions, including DB2 Express-C (no-cost), DB2 Express Edition, DB2 Personal Edition, DB2 Workgroup Server Edition, DB2 Enterprise Server Edition, and DB2 Advanced Enterprise Server Edition. The ideal mix of features for your business would depend upon the features that come with the base product, size of the business, number of users, and other licensing terms. Throughout this article, DB2 z/OS refers to DB2 10 for z/OS; and DB2 LUW refers to DB2 9.7 for Linux, UNIX and Windows. It should also be noted that Linux on System Z servers would run DB2 LUW and not DB2 z/OS.

About subsystems, instances, and databases

You will find differences in the way DB2 server is referred to and accessed on these two platforms. DB2 server is called a subsystem in z/OS and an instance in LUW. DB2 runs its tasks under a set of address spaces in z/OS system (discussed in Memory structures), as a process in Linux and UNIX operating systems, and as a service (unless you started with /D option) in Windows. Each DB2 z/OS server is identified by a four character subsystem identifier (SSID), and each DB2 LUW server is identified by an instance name that takes effect through the DB2INSTANCE environment variable.

What does the term database mean in each platform? A database in DB2 LUW has its own memory region, processes, and recovery logs, while a database in DB2 z/OS is a logical collection of several table spaces and index spaces with very few parameters defined as default for all objects created under it. While each database in DB2 LUW holds its exclusive catalog tables, it is every DB2 subsystem that holds catalog tables.

What about the system objects?

A DSNDB06 database in DB2 z/OS subsystem, and a SYSCATSPACE table space in DB2 LUW will hold all the DB2 catalog tables.

In non-data sharing DB2 z/OS, DSNDB07 is the work-file database. Table spaces that you create in DSNDB07 will be used as storage for processing SQL joins and sorts as well as for temporary tables. In DB2 LUW, a default system temporary table space called TEMPSPACE1 will be available for such DB2 DBM requirements as sorts or joins. You should create one or more USER temporary table spaces for storing data of temporary tables.

When you do not specify the database name while creating a table space or table in DB2 z/OS, DSNDB04 is the default user database for the created object. If you create a table without specifying a table space, DB2 will create an implicit table space for you. In DB2 LUW, assuming you have not created any user table spaces yet, USERSPACE1 will hold the tables you create without specifying the table space explicitly.

DB2 z/OS uses several information required for normal operation in a special database, DSNDB01. Boot Strap Dataset (BSDS) is a z/OS VSAM dataset that will contain some critical information regarding DB2 recovery log dataset names, DB2 checkpoints and DB2 location details. On the other hand, DB2 LUW database stores several information required for normal operation across several operating system files under local database directory.

How will you make your first remote connection?

To access a remote DB2 LUW server, you should catalog the node by specifying DB2 instance and then catalog the target database under that instance at this cataloged node. Listing 1 shows how you can find DB2 server related details on an AIX host. On the other hand, you must catalog a remote DB2 z/OS using its location. Also, you could store host DB2-related information in Database Connection Services (DCS) Directory through CATALOG DCS DATABASE command, as in Listing 2. The location of DB2 z/OS is defined when the DB2 subsystem is installed, which you can find by executing the -DIS DDF command on the DB2 z/OS server, as shown in Listing 3.

The DB2 z/OS subsystem contains information for establishing connections with remote subsystems in a set of catalog tables called Communication Database (CDB). Also, an additional IBM DB2 Connect license is required to remotely connect to DB2 databases located on host systems including z/OS. You can refer to links in the Resources section for more information about IBM DB2 Connect